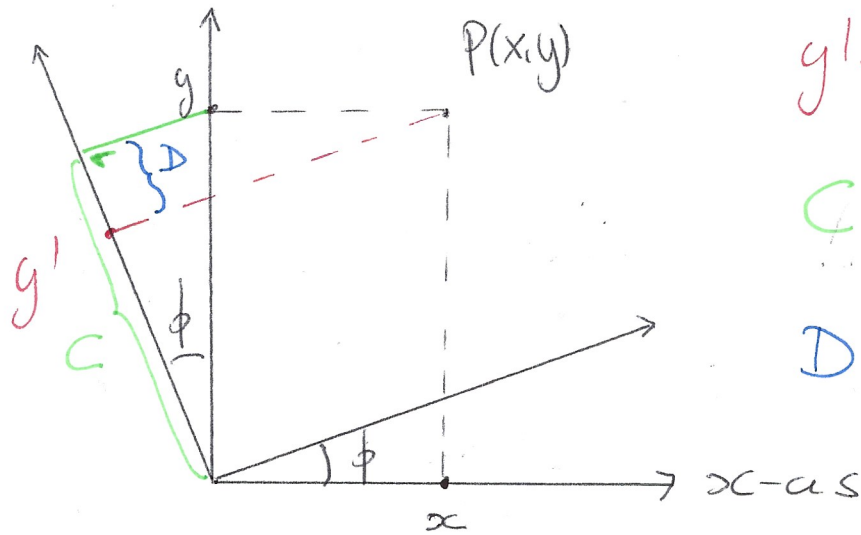
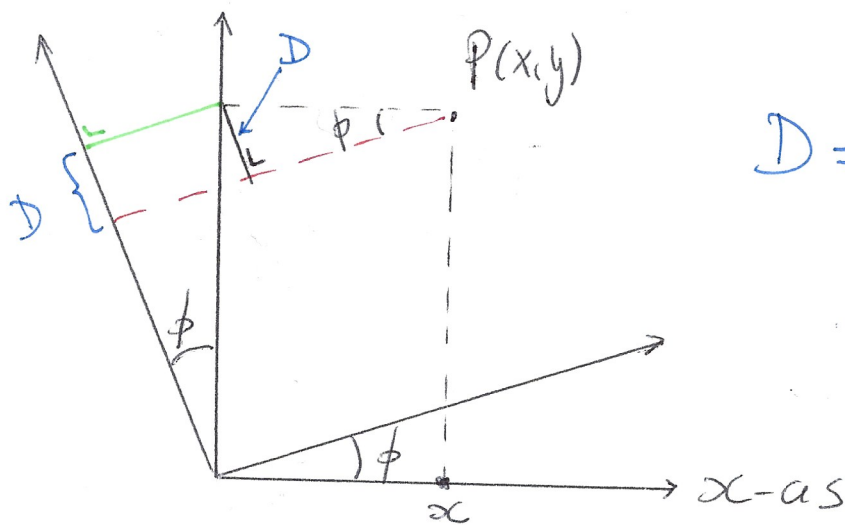




$$y' = C - D$$

$$C = y \cdot \cos \phi$$

$$D = ?$$



$$D = x \cdot \sin \phi$$

$$\Rightarrow y' = -x \sin \phi + y \cos \phi$$

In matrix form:

$$(x', y') = (x, y) \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}$$

2D rotation matrix